

Charles Lin

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2026 – May 2028

Master of Engineering in Electrical and Computer Engineering

Relevant coursework (in progress):

ECE 408 Applied Parallel Programming (CUDA/GPU), CS 432 Operating System Design, ECE 549 Computer Vision

The University of Manchester

Sep 2023 – June 2026

Bachelor of Engineering in Electrical and Electronic Engineering

Ranked 1st in year group

EXPERIENCE

UoM School of Engineering & School of Environment

June 2025 – May 2026

Hardware Design Internship, extended as Final Year Project

Manchester, UK

- Designed a mixed-signal PCB (Altium) for an NDIR methane sensor with 4 Hz crystal-controlled lamp drive, band-pass signal conditioning, partitioned analogue and lamp-drive domains with controlled return currents, and a 16-bit ADC over I²C.
- Wrote embedded firmware for a timer-driven lamp and interrupt-based peak-to-peak sampling of sensor channels, handling detector delay and channel-switch timing; integrated CO₂, pressure, and RTC data into unified timestamped logging.
- Calibrated in two stages: temperature compensation in an environmental chamber and gas-concentration calibration against a commercial analyzer with an automated test/calibration pipeline in Python.
- Validated the £300 platform against a £59k reference instrument: 104 ppm bias, 140 ppm RMSE, 181 ppm 3 σ limit of detection. Concluded it could not resolve peatland fluxes (~6 ppm/h) and identified biogas and mine monitoring as the viable application.

Manchester Stinger Motorsports

Sep 2023 – Nov 2025

Head of Electronics

Manchester, UK

- Participated in the 2025 FSUK competition, first EV entry in history.
- Designed a robust safety reset circuit that allows the shutdown circuit to open or reset based on IMD and BMS conditions.
- Developed the Tractive System Active Light, enabling LEDs to display safety status according to accumulator relays, pre-charge circuitry, and HV presence with non-programmable hardware logic and circuitry.
- Optimized the Brake System Plausibility Device PCB with a 10-second auto-reset circuit, reducing fault recovery time.
- Assisted in recruitment by organizing group assessments and serving as a judge during candidate evaluation.

PROJECTS AND ACTIVITIES

Line Following Robot (Embedded Systems Project)

Sep 2024 – June 2025

- Led a team of 5 to design a high-speed line-following robot using the STM32 microcontroller with C++.
- Developed an algorithm integrating two PID controllers for line following and for motor speed control on slopes.
- Designed the robot chassis in SolidWorks with structural stress analysis and component placement optimization.
- Achieved 6th place out of 50 teams in the university final race competition.

Cache Controller Simulation

April 2025

- Built a direct-mapped cache simulator in C modelling write-back and write-through policies, decoding addresses into tag/index/block fields and reporting hit/miss and memory-traffic statistics from CPU trace files.

Connect 4 Business Card

June 2024

- Designed and built a bare-metal STM32 PCB (KiCad, embedded C) with minimum-system hardware, programmed in C using STM32CubeIDE.
- Posted the project on X, drawing 240K+ views and inbound interest from 10+ prospective buyers.

Peer-Assisted Study Sessions Leader

Sep 2024 – June 2025

- Support first-year engineering students through weekly interactive study sessions.
- Facilitate group discussions, explain difficult concepts, and provide academic mentorship.

AWARDS

University of Manchester

Manchester, UK

- Outstanding Academic Achievement Award - Presented to approximately 0.6% of the student population every year to recognize the top performing students academically as identified by Faculties.
- 1st Prize for Best Overall Performance in the Year Group based on GPA (2024).
- 2nd Prize for Best Overall Performance in the Year Group based on GPA (2025).
- 1st Prize for Best Overall Performance in the Year group based on GPA (2026)

SKILLS

Hardware: Altium, KiCad, mixed-signal PCB design, HV/BMS safety systems, SolidWorks, LTSpice

Embedded: C, C++, STM32, assembly, Mbed, Arduino, I2C, SPI, UART, CAN, PID control

Software & Tools: Python, Git/Github, MATLAB, oscilloscope, multimeter, soldering